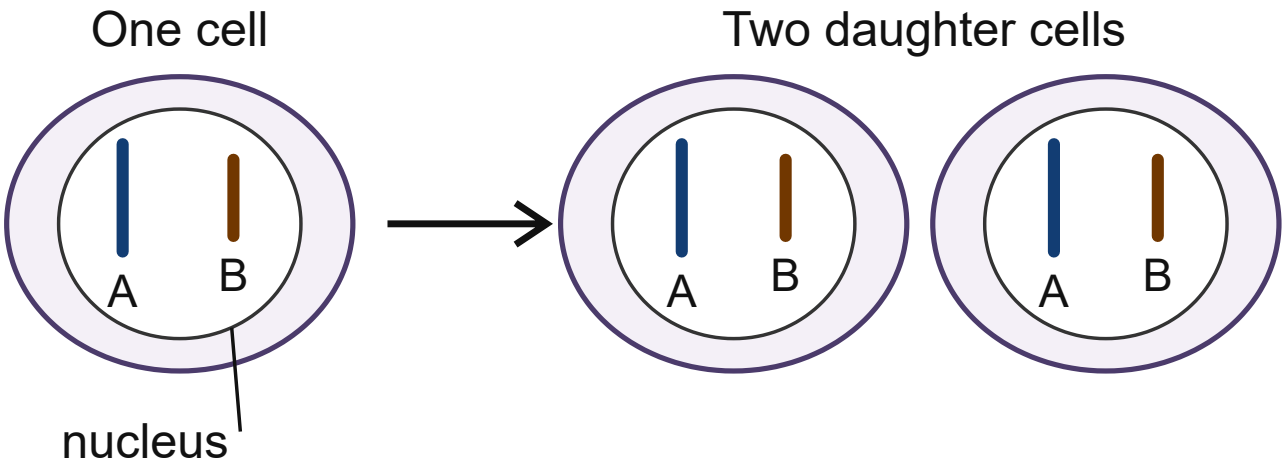
Knowledge organiser



One cell becomes two daughter cells with matching A and B sets.

I am learning to justify the cell-cycle order and explain how matching chromosome sets support growth and repair.

Tier 2 words

Word	Meaning
order	Put steps into a sequence.
matching	Having the same relevant information or parts.
predict	Use what you know to say what might happen.
repair	Replace or mend something damaged.

Tier 3 words

Word	Meaning
cell cycle	The series of steps in which a cell grows and divides.
interphase	The stage in which the cell grows and copies its DNA.
mitosis	The stage in which copied chromosomes separate into two nuclei.
chromosome	A structure of DNA that carries genes.
cytokinesis	The division of the cell after the nuclei form.
prophase	The mitosis phase when chromosomes condense.
metaphase	The mitosis phase when chromosomes line up in the middle.
anaphase	The mitosis phase when the two copies of each chromosome are pulled to opposite ends.
telophase	The mitosis phase when two new nuclei form.

Sequence: interphase → prophase → metaphase → anaphase → telophase → cytokinesis.

How I know: I can order the cell-cycle stages and justify why DNA copying comes first. I can explain how both new body cells receive matching chromosome sets. I can use copy, separate and split to explain how new body cells support growth and repair.

You can plan in any language. Speak before you write: “The daughter cells match because …”

Made by Matt